

TILE TRANSPARENCY SYSTEM

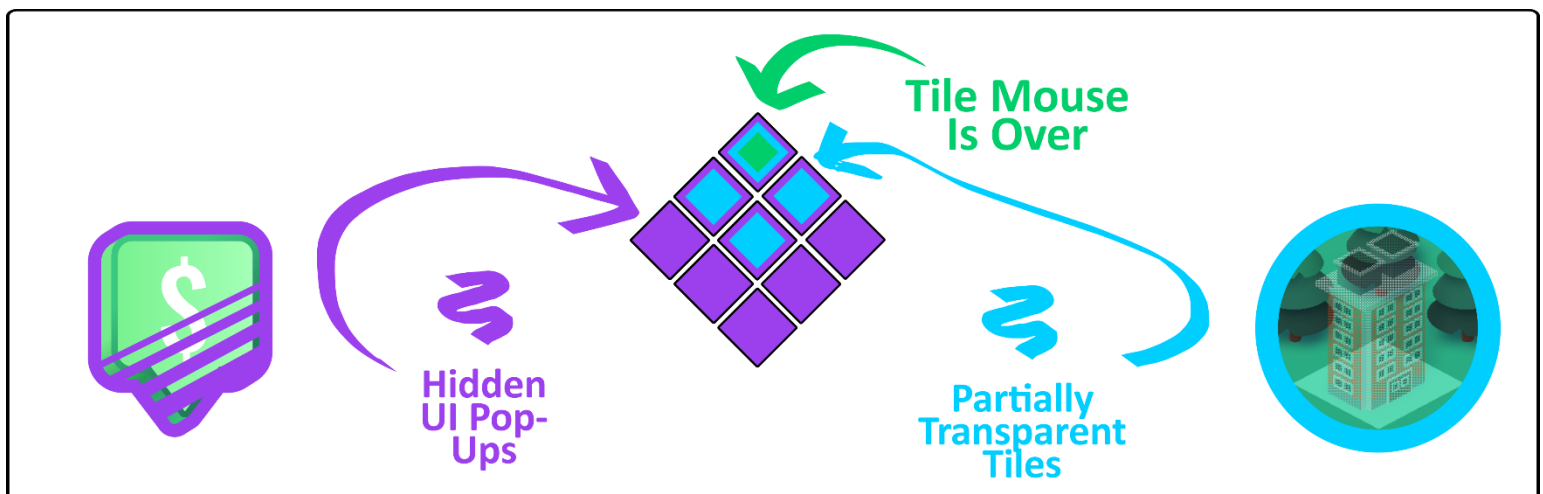
OVERVIEW

The tile transparency system makes tiles partially transparent when the mouse is hovering over them so that it's easier for the player to place objects behind them.

The primary scripts responsible for this system are `MouseHoverTransparency.cs` and `MouseHoverHideTile.cs`. An instance of the `MouseHoverHideTile` class sits on each tile and is in charge of making them partially transparent. The `MouseHoverTransparency` class sits on the Mouse Hover Transparency Manager game object and is in charge of telling tiles under the mouse to hide themselves.

HOW THE SYSTEM WORKS

The `MouseHoverTransparency` class keeps a list of all of the tiles that it currently has partially hidden; the list is called `deactivatedModels`. It also keeps track of every time that the mouse moves to a new grid coordinate. Whenever the mouse moves, it unhides all the models stored in `deactivatedModels` and then hides some of the tiles surrounding the mouse's new grid location. It also hides some of the over-tile UI pop ups, in a slightly larger radius than the transparent tiles.



HOW THE TRANSPARENCY WORKS

A dither transparency shader is used by the tile transparency system to make the tiles partially transparent. A dither transparency shader works by turning off pixels of an object in a pattern, creating the illusion that the object is partially transparent. Here's what it looks like up close:



The incredibly creative name of the dither shader being used for this system is “Dither Shader” which is assigned to all the model’s materials in the Editor. To speed up the process of assigning the shader to all models’ materials, I added a tool in the editor under Tools > Set Object Material Shaders. There should be more information on that in the doc on how to create a new tile.

INTERNAL API

All of the functions of the `MouseHoverTransparency` class are private because it doesn't really need to be interacted with. Instead, I've listed the important public functions of the `MouseHoverHideTile` class, an instance of which sits on each tile.

`MouseHoverHideTile.cs`

`void HideTile(float hoverTransparency)`

Makes the tile partially transparent, at the percentage specified by "hoverTransparency," which should be a number from 0 to 1.

`void UnhideTile()`

Makes the tile fully opaque.

`void HideTileUIPopUps(float hoverTransparencyOfUIPopUps)`

Hides the tile's over-tile UI-Pop ups. Fades them to the transparency specified by "hoverTransparencyOfUIPopUps," which should be a decimal between 0 and 1.

`void UnHideUIPopUps()`

Makes all the over-tile UI-pop ups their normal transparency.

-Graydon Bruyn, May 2026